

RealtouchHR

HMRC RTI Sandbox Testing and Go-Live Implementation Report

Full Payment Submission (FPS) and Employer Payment Summary (EPS)

Prepared For	Purpose	Version	Date
RealtouchHR application team / developer / Codex	To fully test HMRC RTI submission workflow in sandbox/test mode before enabling production/live submissions.	v1.0	08 June 2026

Important: do not paste live Government Gateway credentials into chat tools, source code, tickets, logs, screenshots, or test files. Credentials must be entered only through secure application settings or secret-management tooling.

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1. Executive Summary and Definition of Done

The objective is to make the RealtouchHR HMRC RTI module fully operational in sandbox/test mode before any production/live HMRC submissions are enabled. The application must support Full Payment Submission (FPS) and Employer Payment Summary (EPS) workflows, with clear validation, explicit human approval, safe credential handling, receipt storage, and an auditable submission history.

Area	Definition of Done
Configuration	PAYE reference, Accounts Office reference, employer details, RTI mode, and test credentials can be saved securely and validated without exposing secrets.
Sandbox isolation	Sandbox/test mode cannot send anything to production/live HMRC endpoints. Test credentials and endpoints are visibly separated from production configuration.
FPS	An approved pay run can be selected, validated, prepared as an FPS payload, explicitly approved, submitted to the test endpoint, and recorded with a response/receipt.
EPS	An EPS can be prepared for valid EPS reasons, submitted to the test endpoint, and recorded with a response/receipt.
Audit	Every step is recorded: user, timestamp, company, pay run, submission type, payload hash, submission ID, response status and error details.
Production readiness	The application blocks live mode until all mandatory tests pass and a senior/admin user manually enables production.

Developer summary: **build the safest possible sandbox-first RTI workflow.** The system should automate validation and preparation, but should not remove human approval before submission.

2. Current System State from the Screenshots

The screenshots show that RealtouchHR already has two relevant areas: HMRC RTI Submissions and RTI Sync Engine. The visible system state indicates that test/sandbox mode is active, no approved pay runs are available, and the application is asking for PAYE Reference and Accounts Office Reference to be configured in Settings before RTI submissions can be made.

Screen Item	Meaning	Required Action
Test Mode Active / SANDBOX Mode	The system is correctly positioned for pre-live testing.	Keep this mode active until all sandbox tests pass.
No approved pay runs available	RTI submission cannot proceed until payroll is run and approved.	Create a test pay run, validate calculations, and approve it.

Configuration Required	PAYE Reference and Accounts Office Reference are missing or not mapped.	Add RTI configuration fields in Settings and validate them before submit.
Human approval note	The system already recognises that submission should require explicit approval.	Keep this control in sandbox and production.

3. HMRC RTI Background for Developers

HMRC requires payroll information to be reported electronically through payroll software. An FPS is used to report payments to employees and deductions made from their pay, and an EPS is used where additional employer information or reductions need to be reported. HMRC guidance also states that payroll information is reported by submitting FPS and EPS electronically through the payroll system. See References [1], [2] and [3].

For the technical build, the developer should use HMRC RTI software developer documentation and the relevant tax-year technical specifications. HMRC maintains a Real Time Information support collection and yearly RTI internet submissions technical packs for software developers. See References [4], [5] and [6].

Submission Type	When Used	RealtouchHR Trigger
FPS - Full Payment Submission	Each time employees are paid; normally on or before payday.	Approved pay run exists and contains employee payment/deduction records.
EPS - Employer Payment Summary	When the employer needs to report employer-level adjustments, recoveries, no payment in a period, or similar EPS cases.	Admin selects an EPS reason or the system detects a valid EPS condition.
Final payroll report	At tax year end or when closing payroll/PAYE scheme depending on circumstances.	Year-end module or final submission flag. This must be separately validated.

4. Required Pre-Configuration and Data Inputs

Before sandbox submission can work, RealtouchHR must collect and validate the correct configuration data. Use test data/test credentials during sandbox implementation. Do not use live credentials until production mode is intentionally enabled.

Configuration Field	Purpose	Validation Rule
rti_mode	Controls sandbox/test vs production/live.	Enum only: sandbox, production. Default must be sandbox.
gateway_user_id	HMRC Government Gateway user ID or HMRC test credential	Required for submission. Must be encrypted/secret-managed.

	username.	Never display after save.
gateway_password	HMRC Government Gateway password or HMRC test credential password.	Required for submission. Must be encrypted/secret-managed. Never log.
paye_reference	Employer PAYE reference.	Required. Preserve slash format where applicable. Validate against accepted HMRC pattern before payload generation.
accounts_office_reference	HMRC Accounts Office reference.	Required. Validate length/pattern and strip accidental spaces.
employer_name	Employer legal/trading name.	Required. Should match PAYE scheme record where possible.
employer_address	Employer registered/trading address.	Required for employer metadata where required by payload/schema.
contact_email / contact_phone	Operational contact and support.	Validate format.
vendor_id / sender_id if applicable	Software developer/vendor identity where required by HMRC XML submission route.	Use HMRC developer registration details/test credentials as required by the latest technical pack.

- ☐ Create/confirm Settings screen for HMRC RTI configuration.
- ☐ Add server-side validation for PAYE Reference and Accounts Office Reference.
- ☐ Encrypt or secret-manage HMRC credentials at rest.
- ☐ Mask secrets in UI after save, for example: *****1234.
- ☐ Add a Test Connection / Validate Configuration button that does not submit payroll data.
- ☐ Ensure configuration is company-scoped so one company cannot use another company's credentials.
- ☐ Block FPS/EPS submission if required configuration is missing.

5. Sandbox/Test Mode Implementation Plan

The developer should implement sandbox mode as a strict, isolated environment. It should be impossible for sandbox actions to submit to live HMRC endpoints. The mode must be visible in the UI, logged in the audit trail, and included in every submission record.

Step	Developer Action	Expected Outcome
1	Confirm current RTI code path and identify whether it already generates FPS/EPS payloads or only displays UI.	Clear understanding of what exists and what is missing.
2	Implement environment switch: sandbox endpoint/credentials	Sandbox and production are separated in code, config and

	vs production endpoint/credentials.	database.
3	Add configuration validators for PAYE and Accounts Office references.	The warning shown in the screenshot disappears only when valid configuration exists.
4	Implement local data validation before payload generation.	Missing employee/payroll values are caught before HMRC submission.
5	Generate FPS payload for an approved pay run.	The system can produce a schema-valid test payload.
6	Generate EPS payload for selected EPS reason.	The system can produce a schema-valid EPS test payload.
7	Submit to HMRC test service or HMRC-compatible local stub/mock if official test credentials are not yet available.	The integration can be tested safely without live submissions.
8	Store request metadata, payload hash and HMRC/test response.	Recent Submissions and Audit Log show complete traceability.
9	Run negative tests: bad references, invalid employee data, credential failure, duplicate submission, timeout.	The app handles errors safely and explains how to fix them.
10	Lock production mode behind an admin-only readiness checklist.	No accidental live submission is possible.

Recommended environment variables / secret names

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RTI_MODE=sandbox
HMRC_RTI_SANDBOX_ENDPOINT=<from HMRC technical pack or test service>
HMRC_RTI_PRODUCTION_ENDPOINT=<from HMRC technical pack>
HMRC_RTI_GATEWAY_USER_ID=<stored in secret manager>
HMRC_RTI_GATEWAY_PASSWORD=<stored in secret manager>
HMRC_RTI_VENDOR_ID=<if applicable>
HMRC_RTI_SOFTWARE_NAME=RealtouchHR
HMRC_RTI_SOFTWARE_VERSION=<current app version>

```

6. FPS Workflow: Validate, Prepare, Approve, Submit, Store Receipt

Workflow Stage	Application Behaviour	Developer Requirement
Select	User selects an approved pay run.	Only approved and locked pay runs should appear. Draft pay runs must be hidden or disabled.
Validate	System checks employer configuration, employee data, tax/NI data, pay dates,	Return clear grouped errors: Employer Config, Employee Data, Payroll Calculation,

	deductions and schema-required fields.	Submission Rules.
Prepare	System generates FPS payload for the pay run.	Payload must be generated from immutable approved payroll data, not editable draft fields.
Approve	User sees summary and confirms submission.	Approval must record user ID, timestamp and submission summary.
Submit	System sends to sandbox/test endpoint.	Use retry-safe logic. Do not duplicate successful submissions.
Done	System stores response and receipt.	Show status: accepted, rejected, pending, failed, timeout, retryable, non-retryable.

FPS state machine

DRAFT_PAYRUN -> APPROVED_PAYRUN -> FPS_VALIDATED -> FPS_PREPARED -> FPS_USER_APPROVED -> FPS_SUBMITTED
-> FPS_ACCEPTED

| -> FPS_REJECTED
| -> FPS_PENDING
| -> FPS_FAILED_RETRYABLE
| -> FPS_FAILED_NON_RETRYABLE

Mandatory FPS Validation Checks

- ☐ Company has RTI configuration completed.
- ☐ Sandbox mode is active during testing.
- ☐ PAYE reference and Accounts Office reference are present and valid.
- ☐ Pay run is approved and locked.
- ☐ Pay date exists and belongs to the correct tax year/period.
- ☐ Every paid employee has full name, start date, date of birth, address, tax code, NI category, pay frequency and payroll identifiers as required.
- ☐ Gross pay, taxable pay, tax deducted, employee NI and employer NI reconcile with payroll calculation output.
- ☐ No employee is accidentally duplicated in the same FPS.
- ☐ Submission has an idempotency key to prevent duplicate submission after refresh/retry.
- ☐ Payload validates against the relevant HMRC RTI schema/business rules before transmission.

7. EPS Workflow: Trigger Rules, Validate, Submit, Store Receipt

EPS should not be treated as an automatic monthly submission in every case. It should be prepared only where there is a valid EPS reason. Examples include reporting statutory payment recovery,

claiming Employment Allowance where applicable, reporting no payment in a tax month, or other employer-level adjustments supported by the current HMRC guidance and technical pack.

EPS Scenario	UI Trigger	Validation Requirement
No employees paid in a tax month	Admin selects No Payment / No FPS due reason.	Confirm no FPS/payments exist for the period before allowing EPS.
Statutory payment recovery	System detects recoverable statutory pay or admin enters recovery amount.	Validate amount against payroll records and permitted fields.
Employment Allowance claim	Admin enables/claims if eligible.	Require eligibility confirmation and store declaration/audit.
CIS deductions suffered, if applicable	Admin enters amount where the employer is eligible/applicable.	Validate the company type and amount before payload generation.
Final or ceased indicator where applicable	Year-end/PAYE closure workflow.	Block unless the admin confirms final/cessation intent.

EPS state machine

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EPS_REASON_SELECTED -> EPS_VALIDATED -> EPS_PREPARED -> EPS_USER_APPROVED -> EPS_SUBMITTED ->
EPS_ACCEPTED
                                | -> EPS_REJECTED
                                | -> EPS_FAILED_RETRYABLE
                                | -> EPS_FAILED_NON_RETRYABLE

```

8. Security, Credentials, Audit Trail and Error Handling

RTI integration deals with employer credentials, payroll data and employee information. Treat it as a high-security workflow. The developer should implement security controls before any real employer data is processed.

Risk	Required Control
Credential exposure	Never log passwords. Encrypt at rest. Mask in UI. Restrict access to payroll admins only.
Accidental live submission	Default to sandbox. Require admin-only production switch and readiness confirmation.
Duplicate FPS/EPS	Use idempotency key based on company, pay run, submission type and period. Block resubmission unless explicitly marked as correction/retry.
Rejected submission not fixed	Show actionable HMRC/test error details and keep status open until corrected or superseded.
Untraceable changes	Audit all configuration changes and submission actions.
Payload data leak	Store full payload only if legally/operationally required; otherwise store secure archive with

	access control plus payload hash.
Timeout during submission	Mark as pending/unknown until response/polling confirms accepted or rejected. Do not blindly resubmit.

Audit Field	Required?
audit_id	Yes
company_id	Yes
user_id	Yes
action	Yes - validate, prepare, approve, submit, retry, cancel, config_change
submission_type	Yes - FPS or EPS
pay_run_id / tax_period	Yes where applicable
rti_mode	Yes - sandbox or production
payload_hash	Yes
hmrc_correlation_id / transaction_id	Yes if returned or generated
status and response_code	Yes
error_summary and raw_error_reference	Yes for failures
created_at	Yes

9. Test Scenarios and Acceptance Criteria

The following scenarios should be run before live mode is enabled. Each test should be evidenced with screenshot or log reference, and each successful sandbox submission should appear under Recent Submissions and Audit Log.

Test ID	Scenario	Expected Result	Pass Evidence
T01	Open RTI Sync with missing PAYE/Accounts Office reference.	Submission blocked with clear configuration message.	Screenshot + audit log.
T02	Enter invalid PAYE reference format.	Save blocked or validation error shown.	Screenshot.
T03	Enter valid sandbox configuration.	Configuration saved; Test Connection/Validate passes or reaches configured test endpoint.	Settings screenshot + log.
T04	No approved pay run exists.	RTI screen shows no approved pay runs and submit is disabled.	Screenshot.
T05	Create draft pay run only.	Draft pay run cannot be selected for FPS.	Screenshot.
T06	Approve a single-employee monthly pay	Pay run appears in RTI Select Pay Run list.	Screenshot.

	run with complete data.		
T07	Validate approved pay run.	FPS validation passes; generated validation report stored.	Validation record.
T08	Remove required employee data and retry validation.	Validation fails with exact missing field explanation.	Screenshot + error log.
T09	Prepare FPS in sandbox.	Schema-valid payload generated and payload hash stored.	Payload metadata record.
T10	Approve FPS submission.	User approval timestamp and user ID stored.	Audit log.
T11	Submit FPS to sandbox/test endpoint.	Submission status becomes accepted, rejected, or pending with response retained.	Recent Submissions record.
T12	Retry after timeout.	System does not duplicate submission; uses safe pending/retry logic.	Log evidence.
T13	Submit the same accepted FPS again.	Blocked unless correction/resubmission workflow is explicitly allowed.	Screenshot.
T14	Prepare EPS for no-payment period.	EPS validates and submits in sandbox with receipt stored.	Recent Submissions record.
T15	Attempt EPS where FPS/payments already exist for no-payment reason.	Blocked with clear reason.	Screenshot.
T16	Credential failure test.	System reports authentication/configuration failure without exposing secrets.	Error screenshot.
T17	Switch production toggle as non-admin.	Denied.	Permission log.
T18	Switch production toggle before tests complete.	Blocked by readiness checklist.	Screenshot.
T19	Audit log export.	All RTI events export correctly for compliance review.	CSV/PDF export.
T20	End-to-end sandbox demo.	Developer can demonstrate Select -> Validate -> Prepare ->	Screen recording or signed checklist.

		Approve -> Submit -> Done.	
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10. Go-Live Readiness Checklist

Production/live mode should remain disabled until every item below is completed and signed off. This list should be implemented inside the application as a production gate, not only used manually.

- ☐ All sandbox FPS tests have passed.
- ☐ All sandbox EPS tests have passed.
- ☐ Configuration validation works for PAYE Reference and Accounts Office Reference.
- ☐ Approved pay runs only can be submitted.
- ☐ Human approval is required immediately before every submission.
- ☐ Duplicate prevention/idempotency has been tested.
- ☐ Credential storage is encrypted/secret-managed and secrets are masked in UI.
- ☐ Audit log captures all RTI configuration and submission events.
- ☐ Recent Submissions page shows FPS/EPS status, date, user, company, pay run/period and response.
- ☐ Rejected submissions provide clear error messages and remediation steps.
- ☐ Timeout/pending states are handled without unsafe resubmission.
- ☐ Live endpoint and sandbox endpoint cannot be confused in code/config.
- ☐ Production toggle is admin-only and disabled by default.
- ☐ A final written approval is recorded before first live submission.
- ☐ Live credentials are entered only in secure settings/secret manager, not in code or chat.

11. Codex-Ready Build Prompt

The following can be pasted into Codex or sent directly to the developer. Replace placeholders with the actual repository details and stack.

TASK: Complete RealtouchHR HMRC RTI sandbox/test integration before production go-live.

Context:

- RealtouchHR already has HMRC RTI and RTI Sync screens.
- Current UI shows SANDBOX/Test Mode and blocks submission because PAYE Reference and Accounts Office Reference are not configured.
- Goal is to fully test FPS and EPS workflows in sandbox/test mode before enabling live production HMRC submissions.

Build requirements:

1. Add/complete HMRC RTI Settings for each company:
 - `rti_mode`: sandbox | production, default sandbox
 - PAYE Employer Reference
 - Accounts Office Reference
 - Employer name/address/contact
 - HMRC gateway/test credentials stored securely

- Vendor/software identifiers where required by HMRC RTI XML technical pack
- 2. Ensure credentials are encrypted/secret-managed, masked in UI, and never logged.
- 3. Implement strict environment isolation:
 - Sandbox credentials/endpoints can only submit to sandbox/test service.
 - Production credentials/endpoints can only be enabled after readiness checklist passes.
- 4. Implement FPS workflow:
 - Select approved pay run only
 - Validate employer config, employee data and payroll calculations
 - Generate schema-valid FPS payload for the relevant tax year
 - Require human approval
 - Submit to sandbox/test endpoint
 - Store receipt/response, payload hash and audit record
- 5. Implement EPS workflow:
 - Allow only valid EPS reasons
 - Validate period and employer-level amounts
 - Generate schema-valid EPS payload
 - Require human approval
 - Submit to sandbox/test endpoint
 - Store receipt/response and audit record
- 6. Implement RTI submission state machines:
 - draft/validated/prepared/approved/submitted/accepted/rejected/pending/failed_retryable/failed_non_retryable
- 7. Implement idempotency and duplicate prevention for FPS/EPS.
- 8. Implement detailed errors without leaking secrets or personal data.
- 9. Implement Recent Submissions and Audit Log records for all RTI actions.
- 10. Add automated and manual tests for the 20 test scenarios in the implementation report.
- 11. Block production/live mode until all tests pass and admin confirms readiness.

Acceptance criteria:

- Demo can run end-to-end in sandbox: Select -> Validate -> Prepare -> Approve -> Submit -> Done.
- No draft pay run can be submitted.
- No submission can proceed without PAYE Reference and Accounts Office Reference.
- Duplicate submission is blocked.
- Failed/rejected submission is clearly displayed with remediation guidance.
- Production cannot be enabled by non-admin users or before readiness checks are complete.
- All submission actions are auditable.

Use HMRC's current RTI technical specifications for the relevant tax year and do not invent production endpoint or payload rules. If official HMRC test credentials are not yet available, implement a local HMRC-compatible mock/stub for integration testing, but keep the code structured so the official test service can be connected cleanly.

12. References

[1] GOV.UK - Running payroll: Reporting to HMRC: FPS:

<https://www.gov.uk/running-payroll/reporting-to-hmrc>

[2] GOV.UK - Running payroll: Reporting to HMRC: EPS:

<https://www.gov.uk/running-payroll/reporting-to-hmrc-eps>

[3] GOV.UK - 2025 to 2026 Employer further guide to PAYE and National Insurance contributions:

<https://www.gov.uk/government/publications/cwg2-further-guide-to-payee-and-national-insurance-contributions/2025-to-2026-employer-further-guide-to-payee-and-national-insurance-contributions>

[4] GOV.UK - Real Time Information support for software developers:

<https://www.gov.uk/government/collections/real-time-information-online-internet-submissions-support-for-software-developers>

[5] GOV.UK - Real Time Information internet submissions: generic technical specifications:

<https://www.gov.uk/government/publications/real-time-information-internet-submissions-generic-technical-specifications>

[6] GOV.UK - Real Time Information internet submissions: 2026 to 2027 technical specifications:

<https://www.gov.uk/government/publications/real-time-information-internet-submissions-2026-to-2027-technical-specifications>

[7] GOV.UK - Basic guide for XML software developers: <https://www.gov.uk/guidance/basic-guide-for-xml-software-developers>

[8] GOV.UK - PAYE Online for employers: <https://www.gov.uk/payee-online>

[9] GOV.UK - PAYE Online enrolment: <https://www.gov.uk/payee-online/enrol>

[10] GOV.UK - Register as an employer: <https://www.gov.uk/register-employer>